Safe Smart Maps - UI Design Brief for Stitch

Cal Hacks 13.0 | Version 0.1 | 24 Oct 2026 | Source of truth for behavior: docs/specs/ and docs/agent-context/CONTRACTS.md. This brief is for the person or tool designing the screens. Numbers and place names shown are **sample content** for design only.

1. What we are building

Normal maps show the fastest walking route. Safe Smart Maps shows **three** walking routes for the time the user will actually be on the street: **Most lit & active** (slowest), **Balanced**, and **Fastest** (least lit and active). Each route is scored on open places, lighting, main roads, help points and transit, and, most importantly, on whether those things are spread along the **whole** route. A route with a busy first 20% and an empty last 80% must not look good. Colored segments on the map make weak stretches obvious.

2. Who it is for

Maya, 20, undergrad. Leaves the library at 10:40 PM and walks 15-25 minutes to transit or home, often alone, on her phone, one-handed, tired eyes, low battery. She will trade a few extra minutes for a route that stays lit and has open places along the whole way, and she wants to see *why* a route is recommended. Secondary viewers: hackathon judges watching a 2-minute demo on a laptop projector.

3. Design principles

- **Mobile first** (375 x 812), one-handed, thumb-reachable controls at the bottom. Desktop layout is a bonus: panel on the left, map on the right.
- **Night-friendly:** default dark theme (dark map style), high contrast, large type, no pure-white glare. Light theme optional.
- **Calm, not scary.** No alarming imagery (shields, warning skulls, flashing red). We describe conditions; we do not promise safety.
- **Always show the reasons** and the time the data was fetched. Trust comes from transparency.
- **Color is never the only signal.** Green/yellow/red segments also carry an icon or line pattern and a text label (color-blind safe).
- **Honest about gaps:** missing or old data is shown as "not sure", never a guessed value. Activity is labeled "estimate".

4. Mandatory wording rules (copy on every screen)

Do	Don't
"Better lit and more active"	"Safe" as a promise, "secure", "dangerous", "crime-free"
Route labels: **Most lit & active**, **Balanced**, **Fastest**	Any claim that guarantees protection
"Not sure" when data is missing or stale	Filling blanks with guesses
"Activity estimate" (never a people count)	"12 people nearby"
"Unusual heart rate detected"	Medical claims or diagnosis
Footer: "Explained by AI, may be wrong. In an emergency call 911."	Hiding the AI/emergency note
Heart-rate readings in demos labeled "Simulated"	Showing simulated data as real

5. Visual system (starting point; Stitch may refine)

Item	Spec
Theme	Dark default. Background #0F1720, surface #17222E, text #F2F5F8, muted text #9FB0C0
Segment bands	Better lit/active (green) #2FBF71, Mixed (yellow) #F2B632, Weak (red-orange) #E5603D, Not sure (grey, dashed) #8A97A3. Each also gets an icon: check / dash / minus / question mark
Route identity	Most lit & active = teal accent, Balanced = blue, Fastest = violet (card accents only). Selected route is thick; others thin and semi-transparent
Typography	Clean sans-serif (Inter or similar). Title 24, card title 18, body 15-16, caption 12. Minimum tap target 48 px
Shape	Rounded corners 16 px, bottom sheet with drag handle, subtle shadows, generous spacing
Icons	Simple outline icons: store (open place), lamp (lighting), road, cross (help point), bus (transit), clock, info
Map	Mapbox GL dark style. Start and destination pins, route lines as colored segments (about 50 m each), help points as small cross markers, open places as small dots

6. User flow

Welcome (first time) -> Plan trip -> (optional) Settings -> Fetching live data -> Results: map + three route cards -> Route detail (reasons, gaps, AI explanation) -> optional Time comparison (8 PM vs 1 AM) -> optional Walk mode. Stretch: Heartbeat check setup and alert flow. Always reachable: About this data.

7. Screens needed

Screens 1-7 and 10 are **must have** for the demo. Screens 8-9 are **nice to have / stretch**. For each screen, design the default state plus the listed states.

Screen 1: Welcome / first-run *[Must]*

Purpose: Explain the idea in 10 seconds and set expectations honestly.

Elements:

- App name and one-line promise: "Walking routes for the time you will actually be on the street."
- Three short points with icons: three routes, reasons shown, time-aware
- Honesty card: "We describe conditions, we do not guarantee safety. In an emergency call 911."
- Button "Get started"; link "How scores work" (opens screen 10)
- Location permission explained in plain words: used only during your trip, deleted afterward, never shared unless you choose

States to design: default; location permission denied (offer manual start entry)

Screen 2: Plan trip (home) *[Must]*

Purpose: Enter where from/to and when. This is the main screen.

Elements:

- Full-screen dark map behind a bottom sheet
- From field (default "My location") and To field with search suggestions (geocoding); swap button between them
- Departure time chip: "Leave now" or a date and time picker. Helper text: "Scores use the time you reach each street, not when you leave"
- Quick chips for the demo: "Tonight 8 PM", "Tonight 1 AM"
- Primary button "Show routes"; secondary icon button "Settings"
- Small footer: "Live data covers the demo area: Palace of Fine Arts to Montgomery BART"

States to design: empty; typing with suggestions; outside demo area (friendly "We only have live data for the demo area"); location denied

Screen 3: Settings *[Must]*

Purpose: Let the user tune the result.

Elements:

- Maximum extra minutes: slider 0-20 min (default 10) with live label
- Walking alone / with someone: segmented control
- What matters most: chips (Lighting, Open places, Main roads, Help points, Transit); choose up to 2
- Reset to default; Done
- Note: "Past-incident data is used lightly because it can reflect policing as much as danger."

States to design: default; changed values; reset

Screen 4: Fetching live data *[Must]*

Purpose: Make the live data visible and keep users patient (also proof of live data for judges).

Elements:

- Progress checklist that fills in: "Finding routes", "Checking what's open when you arrive", "Checking lighting and main roads", "Checking transit", "Writing the explanation"
- Each line shows a plain source name (e.g. "Live business listings") and a tick, "slow" or "not available"
- Cancel button

States to design: in progress; slow source ("Still loading places, showing what we have"); one source failed (continues with a warning)

Screen 5: Results: map + three route cards *[Must]*

Purpose: The core screen: compare the three routes.

Elements:

- Map with the selected route drawn as colored segments; the other two faded; start and destination pins
- Bottom sheet (peek, half, full) with three route cards, tab-like selection
- Each card: label, total minutes and "+6 min vs fastest", route score with a short phrase ("Mostly lit and active"), coverage % (e.g. 86%), longest empty stretch (e.g. 120 m near the park), open places count, nearest help point and distance
- Merged case: one card with two badges and the note "Most lit & active and Balanced are the same route"
- "Data fetched at 10:31 PM" badge on the sheet
- Legend: segment colors + icons + "not sure" (dashed grey)
- Buttons: "Details" on each card, "Compare 8 PM vs 1 AM", "Start walk"

States to design: three distinct routes; two merged; no route within extra-time cap (show Fastest only, with explanation); some segments "not sure"

Screen 6: Route detail *[Must]*

Purpose: Show exactly why this route is rated the way it is.

Elements:

- Header: route label, minutes, score
- Continuity block (our key difference): coverage bar with an 80% target marker, longest empty stretch with location note, worst 200 m window score
- Segment strip: horizontal bar of the whole route colored by segment; tap a part to highlight it on the map
- Tapped segment popover: score, band, arrival time at that point, reasons such as "2 open places within 50 m", "lit main road", "no open places nearby"
- Factor breakdown, 8 small bars with weights: open places 25%, lighting 20%, main roads 15%, help points 10%, transit 10%, activity estimate 10%, past incidents 5%, turns and dead ends 5%; activity labeled "estimate"
- AI explanation card in plain language with "Explained by AI, may be wrong"
- Open places list (name, category, closes at, 24 h badge); data source and fetched-at time
- Buttons: "Walk this route", "Back"

States to design: full data; stale data ("older than 2 h" warning); segment confidence unknown

Screen 7: Time comparison: 8 PM vs 1 AM *[Must (demo moment)]*

Purpose: Show that scores change with the clock: shops close, coverage drops, the best route changes.

Elements:

- Two columns (or a swipe/toggle) for the same trip at 8 PM and 1 AM
- Each shows a map thumbnail with colored segments, coverage %, longest gap, open places count
- One-sentence change summary (sample): "At 1 AM coverage drops from 86% to 52% and the best route changes"
- Time slider to scrub between 6 PM and 3 AM, updating the numbers

States to design: same route both times; route changes; second time still loading

Screen 8: Walk mode *[Nice]*

Purpose: Simple guidance while walking.

Elements:

- Large next-turn instruction at top; map follows the user
- Upcoming stretch indicator: "Next 200 m: well lit and active" or "Next 150 m: quiet stretch"
- Small "I'm okay" and "End walk" buttons; a visible "Call 911" button
- Note: location is used only during the walk and deleted after

States to design: on route; off route (offer re-plan); arrived

Screen 9: Heartbeat check (opt-in) *[Stretch]*

Purpose: Optional alert to a trusted contact. Must feel optional, private and non-medical.

Elements:

- 9a Consent: what is read, why, stays on the phone, deleted after the trip; toggle off by default
- 9b Setup: choose contact (name and phone); the contact must agree to receive alerts (status pending/agreed); sensitivity setting
- 9c Device: connect a Bluetooth heart-rate band, or "Use simulated reading" with a permanent **Simulated** badge
- 9d Check-in prompt: "Unusual heart rate detected. Are you okay?" with a big "I'm okay" button and a 30-second countdown ring
- 9e Alert sent: confirms the contact was messaged with live location; "Call 911" button; "Cancel alert" if sent in error
- Footer on all: "Does not replace calling 911."

States to design: consent off/on; contact pending; countdown running; alert sent; cancelled

Screen 10: About this data and how scores work *[Must (small)]*

Purpose: Responsible design and judge credibility.

Elements:

- Plain explanation of the 8 factors and weights, and the continuity rule with a tiny diagram (coverage, longest gap, worst window)
- Data sources list with what each provides and last-updated time (live business listings, OpenStreetMap, 311 outage reports, transit schedules, public incident data)
- Honest limits: activity is an estimate; incident data has low weight and can unfairly flag areas; data can be missing; AI explanations can be wrong
- Privacy: minimal location, deleted after the trip, sharing is opt-in
- Emergency: call 911

States to design: default

8. Reusable components to define

- Route card (3 variants + merged variant), segment legend, coverage bar with target marker, segment strip, factor bar, reason chip, data-fetched-at badge, "not sure" tag, "estimate" tag, "simulated" badge, bottom sheet (peek/half/full), time picker chip, source-status row, warning banner (slow/stale/blocked), empty and error placeholders, primary/secondary/destructive buttons

9. Data each screen can display (nothing else exists)

Item	Fields
Route	label, also_labels, duration_min, extra_min, distance_m, route_score, open_places_count, nearest_help_point{name, distance_m}, explanation, explanation_source (ai or template)
Continuity	coverage_pct, longest_gap_m, longest_gap_note, worst_window_score
Segment	segment_id, start, end, length_m, arrival_time, score, band (green/yellow/red/unknown), factors, reasons[], data_fetched_at, confidence (high/low/unknown)
Response	api_version, scoring_version, data_fetched_at, warnings[]

Item	Fields
Not available	Live people counts, crime predictions, a guaranteed safety rating. Do not design UI for these.

10. Sample content for mockups

- Trip: Palace of Fine Arts to Montgomery BART, leaving 10:40 PM, about 3.5 km.
- Most lit & active: 46 min (+5), score 78, coverage 88%, longest gap 110 m near the Presidio edge, 14 open places, nearest help point 90 m.
- Balanced: 43 min (+2), score 66, coverage 74%, longest gap 220 m near the park, 9 open places.
- Fastest: 41 min, score 49, coverage 52%, longest gap 380 m, 5 open places.
- Reason examples: "2 open places within 50 m", "lit main road", "bus stop with service in the next 10 min", "no open places for 220 m", "not sure: lighting data missing".
- Mark all these numbers as sample in the files you hand over.

11. Deliverables requested from Stitch

- Mobile screens 1-7 and 10 (and 8-9 if time allows) in the dark theme, with the listed states.
- One desktop layout of screen 5 (results) for the demo projector.
- A one-page component sheet (colors, type, route card, legend, badges).
- Exportable code or images developers can use. The frontend is Next.js with Mapbox GL JS, so keep the map as a full-bleed layer with overlays on top.

12. Paste-ready prompt for Stitch

Design a mobile-first, dark-theme web app called Safe Smart Maps. It compares three walking routes (Most lit & active, Balanced, Fastest) for a late-night student, scoring each on open places, lighting, main roads, help points and transit, and on continuity along the whole route (coverage %, longest empty stretch, worst 200 m window). Never use the word safe as a promise; use better lit and more active, and always show reasons, a data-fetched-at time, and a not sure state. Screens: Welcome, Plan trip (from/to, departure time, quick chips 8 PM and 1 AM), Settings (extra minutes slider, alone or with someone, priorities), Fetching live data checklist, Results (full-bleed map with segments colored green/yellow/red plus icons and patterns, bottom sheet with three route cards), Route detail (continuity block, segment strip, factor bars, AI explanation, open places), Time comparison 8 PM vs 1 AM, About this data. Stretch: Walk mode and an opt-in heartbeat check flow (consent, contact, check-in countdown, alert sent) with a visible Simulated badge. Calm tone, large tap targets, accessible contrast, color never the only signal.

13. Constraints and notes

- Scope is one demo area, not a whole city; do not design city pickers.
- No accounts or login screens. No payment screens.
- Anything not listed in section 9 does not exist in the data; ask before adding it.
- Questions or contract changes go to Bhavpreet or Amey. Shared files: docs/agent-context/CONTRACTS.md and docs/specs/FUNCTIONAL_SPEC.md.